

Chao Ouyang

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EDUCATION

Wuhan University (Ranked #85 Globally, U.S. News 2026–2027)

Sep. 2024 – Jun. 2027 (expected)

Master of Science in Control Science and Engineering

Wuhan, China

- Advisor: Prof. Jun Zhang (recommended for admission without entrance examination)

Wuhan University

Sep. 2020 – Jun. 2024

Bachelor of Science in Automation

Wuhan, China

- GPA: 3.83/4.00 (ranked 9/90, top 10%)

RESEARCH INTERESTS

- **LLM-based Multi-Agent Systems (MAS)**: how agents communicate, how they are organized, and how to optimize the MAS as a whole rather than simply agent by agent.
- **Representation Learning**: how to learn more discriminative representations, mainly through contrastive learning.

PUBLICATIONS

* CORRESPONDING AUTHOR, S = IN SUBMISSION, C = CONFERENCE, J = JOURNAL

[S.1] Chao Ouyang, Qingyuan Liu, Haozhen Zhang, Yuyang Bai, Wenya Wang, and Jun Zhang. **Training Skillful Orchestrators for Dynamic Workflows by Separating Skill from Luck.**

In submission to *ACL Rolling Review (ARR)*. 2026.

[C.1] Chao Ouyang, Yuyang Bai, Jun Zhang*, Tianlu Gao, Yuxin Dai, Peidong Xu, and David Wenzhong Gao. **CacheMAS: Cache Communication for Single-Pass, Jointly-Optimized Multi-Agent Systems.**

In *Advances in Neural Information Processing Systems (NeurIPS)*. 2026.

[C.2] Chao Ouyang, Yuyang Bai, Jun Zhang*, Tianlu Gao, Jun Hao, Lijun Kong, and David Wenzhong Gao. **DisCo DETR: Distance-aware Multi-view Contrastive Learning for DETR Pre-training.**

In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*. 2026. [Paper] [Code]

[J.1] Chao Ouyang, Jie Zhang, Yuyang Bai, Yan Xu, Jun Zhang*, Yun Su, Tianlu Gao, and Yuxin Dai. **Cluster-Awareness and Fine-Grained Feedback Driven Optimization Method for LLM-based Multi-Agent Systems.**

Computer Engineering and Applications. 2025.

RESEARCH EXPERIENCE

GLINT Lab, College of Computing and Data Science, Nanyang Technological University

Jun. 2026 – Present

Research Intern (Remote) advised by Prof. Wenya Wang

Singapore

- **Training Skillful Orchestrators for Dynamic Workflows by Separating Skill from Luck (Lead Researcher)**: Built an orchestrator that drafts a global workflow, instantiates its steps as parallel subagent calls, and revises it at runtime from their results. Trained its delegation decisions with an advantage decomposition that separates the orchestrator's skill from the luck of subagent returns and removes the luck term. Outperformed prior methods by 4–5% in all three domains. **First-author paper in submission to ARR.**

School of Electrical Engineering and Automation, Wuhan University

Sep. 2023 – Present

Research Assistant advised by Prof. Jun Zhang

Wuhan, China

- **CacheMAS: Single-Pass Multi-Agent Systems (Lead Researcher)**: Found that the prefill KV cache alone suffices as the inter-agent signal and compressed a multi-agent pipeline into one prefill and one decode, so that sequence-level RL jointly optimizes all agents without per-agent rewards. Versus text-based multi-agent systems: $\sim 3\times$ fewer tokens, $\sim 2\times$ faster inference, $\sim 6\times$ faster RL training, +1 to +8 points after joint optimization on math, multi-hop QA, and code. **First-author paper accepted at NeurIPS 2026.**
- **CAFG: Cluster-Aware Optimization of LLM Multi-Agent Systems (Lead Researcher)**: Automated multi-agent system design: merged redundant search branches via Louvain clustering over topology, call-sequence, and semantic features (design redundancy 63% $\rightarrow$ 54%), and turned the sparse end-to-end reward into node-level feedback that credits each agent. Averaged 84.2 across six benchmarks, +3.0 over the strongest baseline. **First-author paper accepted at Computer Engineering and Applications.**
- **DisCo DETR: Self-Supervised DETR Pre-training (Lead Researcher)**: Reused DETR's bipartite matching as the contrastive signal and fused spatially close multi-view object features into object queries, with no extra matching network or architectural change. Achieved state-of-the-art results among DETR pre-training methods across five DETR variants on COCO and PASCAL VOC ($\sim +1.5$ AP over the strongest baseline on average). **First-author paper accepted at AAAI 2026.**

INDUSTRY EXPERIENCE

Baidu Inc.

Jul. 2025 – Sep. 2025

Algorithm Intern

Beijing, China

- **Highlight-Clip Generation**: Built a pipeline that edits drama episodes into vertical split-screen character clips; an asynchronous multi-GPU producer-consumer redesign made it $\sim 24\times$ faster (~ 4 h $\rightarrow$ ~ 10 min per episode), with 88.7% usable clips.
- **Subtitle Erasure and Translation**: Built a pipeline that erases burned-in subtitles and translates short dramas (OCR-heatmap localization, LLM-corrected OCR, STTN inpainting, context-aware translation), with 85.7% erasure success on 50 ten-minute dramas.

HONORS AND AWARDS

First Prize, 16th Electrician Cup National Mathematical Contest in Modeling

2025

2nd Place Internationally, Yale Spider 1.0 Text-to-SQL Challenge (85.6% execution accuracy)

2023

First Prize in two tracks, National University Students' Smart Car Competition

2022

First-Class Scholarship, Wuhan University

2021

ADDITIONAL INFORMATION

Languages: Mandarin (native), English (professional working proficiency)

Programming: Python, PyTorch, Linux, Git, $\LaTeX$

Frameworks: verl, vLLM; distributed training (TP / DDP / FSDP)